

POFADDER, South Africa

WMO: 684160

Lat: 29.1236S

Lon: 19.3889E

Elev: 982

StdP: 90.07

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.4	3.7	-14.2	1.2	18.9	-12.6	1.4	19.0	8.2	11.1	7.1	9.6	2.3	140	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	36.6	16.5	35.4	16.2	34.2	15.8	19.9	28.0	19.2	28.4	18.5	28.3	3.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	14.3	23.6	16.2	13.0	23.8	14.7	11.8	23.8	61.7	27.9	59.2	28.1	56.5	28.4	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.3	7.3	6.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-1.7	21.1	0.9	0.8	-2.3	21.6	-2.8	22.1	-3.4	22.5	-4.0	23.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.7	26.4	25.9	24.6	20.7	16.9	12.4	11.8	13.9	17.0	20.3	22.4	24.9
	DBStd	6.30	3.14	2.95	3.64	3.96	3.62	3.35	3.63	4.13	4.05	4.44	4.35	3.38
	HDD10.0	51	0	0	0	0	2	13	24	11	1	0	0	0
	HDD18.3	751	0	0	4	23	70	179	205	149	73	32	15	1
	CDD10.0	3602	507	445	453	321	216	85	79	131	211	321	373	461
	CDD18.3	1262	249	212	199	93	26	1	2	11	34	94	138	204
	CDH23.3	13938	3061	2434	2032	706	183	9	11	106	332	984	1562	2518
CDH26.7	6454	1633	1217	898	212	25	0	0	21	93	395	700	1260	
Wind	WSAvg	3.1	3.7	3.3	2.9	2.7	2.4	2.6	2.6	2.8	3.1	3.6	3.9	3.8
Precipitation	PrecAvg	113	13	21	18	18	7	5	5	3	3	6	7	7
	PrecMax	236	43	123	58	154	44	31	56	20	28	35	62	54
	PrecMin	29	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	55	14	26	16	34	11	7	11	5	6	8	16	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	37.5	35.9	32.6	29.3	24.6	24.5	29.4	31.9	35.5	35.9	37.5
		MCWB	16.6	17.7	16.9	15.3	13.6	11.7	10.9	12.5	13.7	15.2	14.9	16.1
	2%	DB	36.9	35.7	34.3	30.9	27.2	22.3	22.8	26.2	29.2	32.8	34.2	35.7
		MCWB	16.4	17.1	16.4	15.5	12.9	10.5	10.0	11.0	12.2	13.8	14.4	15.8
	5%	DB	35.5	34.4	33.0	29.1	25.5	20.6	20.8	23.9	27.2	30.8	32.5	34.2
		MCWB	16.2	16.8	16.0	14.7	12.3	9.7	9.2	10.2	11.6	13.2	14.0	15.4
	10%	DB	33.9	33.0	31.5	27.3	23.7	18.6	18.6	21.5	24.9	28.7	30.8	32.7
		MCWB	15.6	16.4	15.5	14.0	11.9	9.1	8.4	9.2	10.7	12.4	13.6	14.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	21.5	19.8	17.9	16.1	13.0	11.7	13.8	15.2	17.5	17.9	19.2
		MCDB	28.9	29.6	29.2	26.7	21.6	19.7	21.9	22.9	28.6	29.1	28.8	29.2
	2%	WB	19.9	20.2	18.9	17.0	14.5	11.7	10.7	12.2	13.3	15.7	16.4	18.5
		MCDB	27.5	28.3	29.4	25.6	22.9	19.2	21.0	23.7	26.5	27.5	27.9	29.0
	5%	WB	19.1	19.4	18.2	16.2	13.6	10.8	9.8	11.2	12.0	14.2	15.4	17.5
		MCDB	27.6	28.4	28.1	25.3	22.0	18.2	19.3	22.4	25.6	28.5	28.3	29.0
	10%	WB	18.0	18.6	17.4	15.4	12.5	9.9	9.0	9.8	11.0	13.1	14.4	16.4
		MCDB	28.7	28.3	27.8	24.6	22.5	17.2	17.2	20.1	24.2	27.4	28.6	29.1
Mean Daily Temperature Range	5% DB	MDBR	13.8	13.1	12.2	11.4	11.6	10.8	11.7	12.4	13.2	13.5	13.6	14.1
		MCDBR	14.7	14.0	12.5	12.0	12.8	12.6	13.5	14.3	14.8	14.8	14.7	14.9
		MCWBR	5.1	5.1	4.6	4.6	5.6	5.7	6.3	6.4	6.2	5.2	5.3	5.1
	5% WB	MCDBR	11.5	11.1	10.1	10.4	10.5	10.9	12.3	13.1	13.8	13.6	13.0	12.1
		MCWBR	4.6	4.9	4.4	4.3	4.9	5.1	5.9	5.8	6.0	4.9	5.7	4.9
Clear-Sky Solar Irradiance	taub		0.312	0.324	0.314	0.294	0.269	0.257	0.260	0.288	0.308	0.320	0.295	0.299
	taud		2.496	2.474	2.488	2.539	2.588	2.583	2.561	2.438	2.359	2.359	2.503	2.514
	Ebn at Noon		1029	999	977	951	934	924	935	942	967	991	1040	1046
	Edh at Noon		115	115	106	91	78	73	78	99	119	127	114	114
All-Sky Solar Radiation	RadAvg		8.56	7.73	6.64	5.25	4.16	3.64	3.93	4.88	6.18	7.45	8.47	8.91
	RadStd		0.31	0.29	0.21	0.15	0.14	0.13	0.11	0.16	0.21	0.19	0.26	0.29

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page